BITS Pilani
Pilani Campus

CHEM F111 : General Chemistry

Semester II: AY 2023-24

Lecture- 09, 2nd February 2024, Friday

Last Class:

- Hydrogen atom: Discussion on radial part of wavefunction - the Laguerre polynomial; understanding of n;
- Nodes and radial nodes
- Probability and radial distribution function
- Most probable radius; average radius

Orbitals with zero angular momentum

The wavefunctions corresponding to l=0 are called s-type orbitals. They are independent of of θ and φ. In other words, the angular wavefunction is constant.

For example,

$$\Psi_{1s} = Me^{-Zr/a_0}$$

$$\Psi_{2s} = M\left(2 - Zr/a_0\right)e^{-Zr/2a_0}$$

$$\Psi_{3s} = M\left(27 - 18(Zr/a_0) + 2(Zr/a_0)^2\right)e^{-Zr/3a_0}$$

$$\Psi_{4s} = M\left(192 - 144(Zr/a_0) + 24(Zr/a_0)^2 - (Zr/a_0)^3\right)e^{-Zr/4a_0}$$

Note: In all the orbital expressions, the normalization constant, M is distinct.

Orbitals with angular momentum vector lying in xy plane

These orbitals are characterized by m_l=0 by virtue of zero z-component of the angular momentum. These orbitals are independent of the azimuthal angle, φ. Excluding the s-orbitals (i.e., l=0, since, already discussed), these are characterized by global maxima of probability density along +z and -z directions and are symmetric with respect to rotation about the z-axis.

For example,

$$\Psi_{2p_z} \text{ or } \Psi_{2p_0} = M\mathbf{r}e^{-Zr/2a_0}\cos\theta$$

$$\Psi_{3p_z} \text{ or } \Psi_{3p_0} = M\mathbf{r}\left(6 - Zr/a_0\right)e^{-Zr/3a_0}\cos\theta$$

$$\Psi_{4p_z} \text{ or } \Psi_{4p_0} = M\mathbf{r}\left(80 - 20(Zr/a_0) + (Zr/a_0)^2\right)e^{-Zr/4a_0}\cos\theta$$

Note: In all the orbital expressions, M is normalization constant and is different for different orbitals.

Orbitals with angular momentum vector lying in xy plane

These orbitals are characterized by m_l=0 by virtue of zero z-component of the angular momentum. These orbitals are independent of the azimuthal angle, φ. Excluding the s-orbitals (i.e., l=0, since, already discussed), these are characterized by global maxima of probability density along +z and -z directions and are symmetric with respect to rotation about the z-axis.

For example,

$$\Psi_{3d_{z^2}} \text{ or } \Psi_{3d_0} = M\mathbf{r^2} e^{-Zr/3a_0}(3\cos^2\theta - 1)$$

$$\Psi_{4d_{z^2}} \text{ or } \Psi_{4d_0} = M\mathbf{r^2}\,(12 - Zr/a_0)\, e^{-Zr/4a_0}(3\cos^2\theta - 1)$$

$$\Psi_{4f_{z^3}} \text{ or } \Psi_{4f_0} = M\mathbf{r^3} e^{-Zr/4a_0}(5\cos^3\theta - 3\cos\theta)$$

Note: In all the orbital expressions, M is normalization constant and is different for different orbitals.

Orbitals with non-zero z-component of orbital angular momentum

These orbitals are characterized by $l>0$, $|m_l|>0$. If one probes to measure the exact value of z-component of the angular momentum in addition to its magnitude, the orbitals are complex with $\Phi_{m_l}(\varphi) = e^{im_l\varphi}$

For example,

$$\Psi_{2p_{\pm1}} = M\mathbf{r}e^{-Zr/2a_0}\mathbf{sin}\theta\mathbf{e}^{\pm\mathbf{i}\phi}$$

$$\Psi_{3d_{\pm1}} = M\mathbf{r}^2e^{-Zr/3a_0}\mathbf{sin2}\theta\mathbf{e}^{\pm\mathbf{i}\phi}$$

$$\Psi_{3d_{\pm2}} = M\mathbf{r}^2e^{-Zr/3a_0}\mathbf{sin}^2\theta\mathbf{e}^{\pm\mathbf{2i}\phi}$$

Note: In all the orbital expressions, M is normalization constant and is different for different orbitals.

Orientation-instructive orbitals versus L_z-eigenstates

Unlike in other orbtitals, when considering orbitals with $l>0$, $m_l \neq 0$, which are the L_z-eigenstates, the regions of probability density maxima are not determninate. However, one can take appropriate linear combinations of the pair of orbitals corresponding to $(n, l, \pm m_l)$, to get a pair of orbitals, for which the probability density maxima can be easily located, at the cost of the z-component of the angular momentum – the sign of the z-component of the angular momentum becomes uncertain.

For example,

$$\Psi_{2p_x} = M\mathbf{r}e^{-Zr/2a_0}\mathbf{sin}\theta\mathbf{cos}\phi$$

$$\Psi_{2p_{\pm 1}} = M\mathbf{r}e^{-Zr/2a_0}\mathbf{sin}\theta\mathbf{e}^{\pm \mathbf{i}\phi}$$

$$\Psi_{2p_y} = M\mathbf{r}e^{-Zr/2a_0}\mathbf{sin}\theta\mathbf{sin}\phi$$

Orientation-instructive orbitals versus L_z-eigenstates

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For example,

$$\Psi_{3d_{xz}} = M\mathbf{r}^2 e^{-Zr/3a_0}\mathbf{sin2\theta cos\phi}$$

$$\Psi_{3d_{\pm 1}} = M\mathbf{r}^2 e^{-Zr/3a_0}\mathbf{sin2\theta e^{\pm i\phi}}$$

$$\Psi_{3d_{yz}} = M\mathbf{r}^2 e^{-Zr/3a_0}\mathbf{sin2\theta sin\phi}$$

Orientation-instructive orbitals versus L_z-eigenstates

Unlike in other orbtitals, when considering orbitals with l>0, m_l≠0, which are the L_z-eigenstates, the regions of probability density maxima are not determninate. However, one can take appropriate linear combinations of the pair of orbitals corresponding to (n, l, $\pm m_l$), to get a pair of orbitals, for which the probability density maxima can be easily located, at the cost of the z-component of the angular momentum – the sign of the z-component of the angular momentum becomes uncertain.

For example,

$$\Psi_{3d_{x^2-y^2}} = M\mathbf{r^2}e^{-Zr/3a_0}\mathbf{sin^2}\theta\mathbf{cos2}\phi$$

$$\Psi_{3d_{\pm 2}} = M\mathbf{r^2}e^{-Zr/3a_0}\mathbf{sin^2}\theta\mathbf{e^{\pm 2i\phi}}$$

$$\Psi_{3d_{xy}} = M\mathbf{r^2}e^{-Zr/3a_0}\mathbf{sin^2}\theta\mathbf{sin2}\phi$$

Linear combinations

Linear combination of complex wavefunctions to get real functions.

<u>Linear combination:</u>

- **Consider two functions $f_1(x)$ and $f_2(x)$; then the combination**

$$f_3(x) = a_1 f_1(x) \pm a_2 f_2(x)$$

is a linear combination of these two functions $f_1(x)$ and $f_2(x)$. a_1 and a_2 are constants.

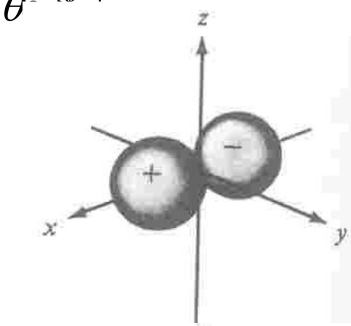
If $f_1(x)$ and $f_2(x)$ are solutions of SE then $f_3(x)$ will also be a solution to the SE.

Real wavefunctions could be defined by linear combinations of complex wavefunctions such that now the maximum probability density maxima could be easily identified.

$$\psi_{2p_1} = R_{2,1}(r)Y_{1,1}(\theta,\phi) \qquad \frac{1}{4(6)^{1/2}}\left(\frac{Z}{a_0}\right)^{3/2}\rho e^{-\rho/4}\left(\left\{\frac{3}{8\pi}\right\}^{\frac{1}{2}}\sin\theta e^{i\phi}\right)$$

$$\psi_{2p_{-1}} = R_{2,1}(r)Y_{1,-1}(\theta,\phi) \qquad \frac{1}{4(6)^{1/2}}\left(\frac{Z}{a_0}\right)^{3/2}\rho e^{-\rho/4}\left(\left\{\frac{3}{8\pi}\right\}^{\frac{1}{2}}\sin\theta e^{-i\phi}\right)$$

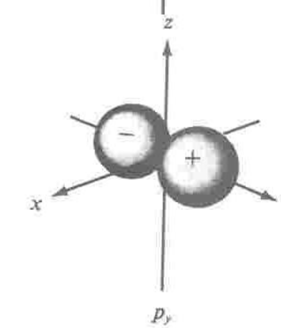
$$\psi_{2p_1} + \psi_{2p_{-1}} = \left\{\frac{1}{\sqrt{16\pi}}\right\}\left(\frac{z}{a_0}\right)^{\frac{5}{2}} e^{\frac{-zr}{2a_0}} r\sin\theta\cos\phi = \psi_{2p_x}$$



1 nodal surface (yz plane)

x= r sin θ cos ϕ

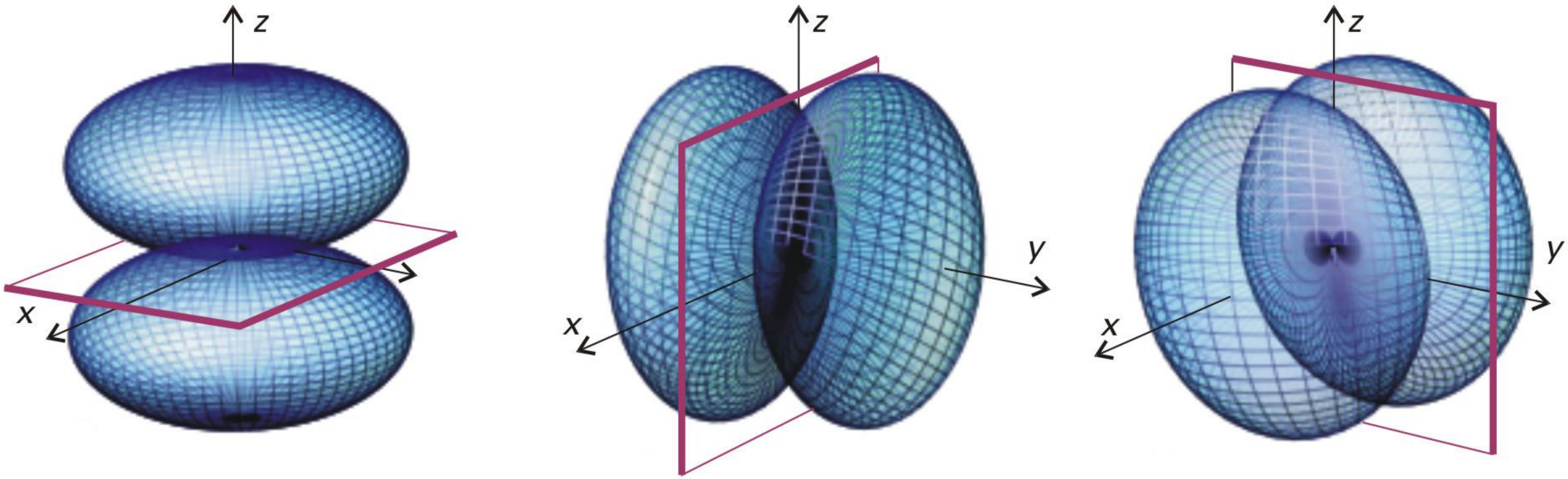
$$\frac{\psi_{2p_1} - \psi_{2p_{-1}}}{i} = \left\{\frac{1}{\sqrt{16\pi}}\right\}\left(\frac{z}{a_0}\right)^{\frac{5}{2}} e^{\frac{-zr}{2a_0}} r\sin\theta\sin\phi = \psi_{2py}$$



1 nodal surface (xz plane)

y = r sin θ sin ϕ

2p orbitals – boundary surfaces



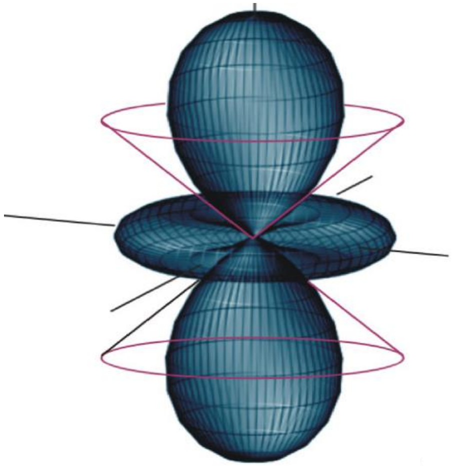
ψ_{2p_x} and ψ_{2p_y}

No. of Radial Node: *n*-*l*-1
No. of angular Node: *l*
Total no. of nodes: (*n*-1)

These non-imaginary wavefunctions are not the eigenfunctions of L_z
(m_l is uncertain in terms of sign).
but are still the eigenfunctions of energy and total angular momentum.

3 d_{z^2} orbitals – boundary surfaces

$$\psi_{3,2,0} = \psi_{3d_{z^2}} = \frac{1}{486}\left(\frac{6Z^3}{\pi a_0^3}\right)^{\frac{1}{2}} \frac{z^2 r^2}{a_0^2} e^{\frac{-zr}{3a_0}} \left(3\cos^2\theta - 1\right)$$

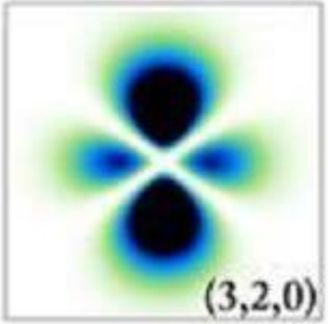


n = 3, l = 2, m$_l$ = 0

No. of radial nodes (n-*l*-1) = 0

No. of nodes (n-1) = 2

No. of Angular nodes *l* = 2



$$= \frac{1}{486}\left(\frac{6Z^3}{\pi a_0^3}\right)^{\frac{1}{2}} \frac{z^2}{a_0^2} e^{\frac{-zr}{3a_0}} \left(3r^2\cos^2\theta - r^2\right)$$

z = r cos θ

$$= \frac{1}{486}\left(\frac{6Z^3}{\pi a_0^3}\right)^{\frac{1}{2}} \frac{z^2}{a_0^2} e^{\frac{-zr}{3a_0}} \left(3z^2 - r^2\right)$$

Other 3d orbitals $(m_l \neq 0)$

$$\psi_{+1} = \frac{1}{81}\left\{\frac{z^3}{\pi a_0^3}\right\}^{\frac{1}{2}}\left(\frac{z^2 r^2}{a_0^2}\right) e^{\frac{-zr}{3a_0}}.\sin\theta\cos\theta.e^{i\phi}$$

$$\psi_{-1} = \frac{1}{81}\left\{\frac{z^3}{\pi a_0^3}\right\}^{\frac{1}{2}}\left(\frac{z^2 r^2}{a_0^2}\right) e^{\frac{-zr}{3a_0}}.\sin\theta\cos\theta.e^{-i\phi}$$

n = 3, l = 2, m$_l$ = =1 or -1

No. of radial nodes (n-*l*-1) = 0

No. of nodes (n-1) = 2

No. of Angular nodes *l* = 2

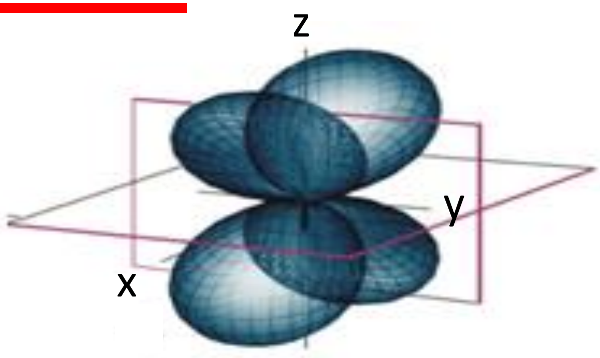
- **These are not real wave functions.**
- **In which direction, the maximum probability density will be oriented ?**

$3d_{xz}$ & $3d_{yz}$ orbitals – boundary surfaces

$$\psi_{3d_1}+\psi_{3d_{-1}}=\left\{\frac{1}{81\sqrt{\pi}}\right\}\left(\frac{z}{a_0}\right)^{\frac{7}{2}}e^{\frac{-zr}{3a_0}}r^2\sin 2\theta\cos\phi$$

$$\psi_{3d_1}+\psi_{3d_{-1}}=\left\{\frac{2}{81\sqrt{\pi}}\right\}\left(\frac{z}{a_0}\right)^{\frac{7}{2}}e^{\frac{-zr}{3a_0}}.\underbrace{r\sin\theta\cos\phi}_{x}.\underbrace{r\cos\theta}_{z}$$

$$=\psi_{3d_{xz}}$$

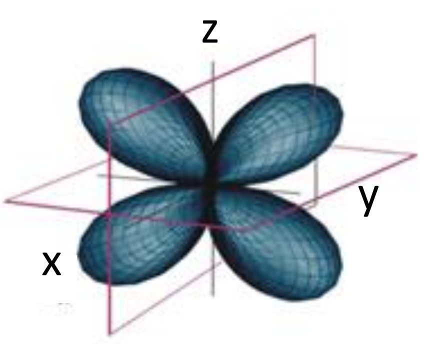


No. of angular Nodes: *l = 2*

$$\frac{\psi_{3d_1}-\psi_{3d_{-1}}}{i}=\left\{\frac{1}{81\sqrt{\pi}}\right\}\left(\frac{z}{a_0}\right)^{\frac{7}{2}}e^{\frac{-zr}{3a_0}}r^2\sin 2\theta\sin\phi$$

$$\frac{\psi_{3d_1}-\psi_{3d_{-1}}}{i}=\left\{\frac{2}{81\sqrt{\pi}}\right\}\left(\frac{z}{a_0}\right)^{\frac{7}{2}}e^{\frac{-zr}{3a_0}}.\underbrace{r\sin\theta\sin\phi}_{y}.\underbrace{r\cos\theta}_{z}$$

$$=\psi_{3d_{yz}}$$

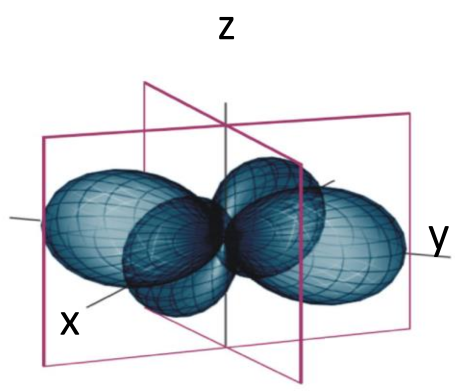


$3d_{x^2-y^2}$ & $3d_{xy}$ orbitals – boundary surfaces

$$\psi_{3d_2}+\psi_{3d_{-2}}=\left\{\frac{1}{81\sqrt{2\pi}}\right\}\left(\frac{z}{a_0}\right)^{\frac{7}{2}}e^{\frac{-zr}{3a_0}}r^2\sin^2\theta\cos 2\phi$$

$$=\left\{\frac{1}{81\sqrt{2\pi}}\right\}\left(\frac{z}{a_0}\right)^{\frac{7}{2}}e^{\frac{-zr}{3a_0}}r^2\sin^2\theta\left(\cos^2\phi-\sin^2\phi\right)$$

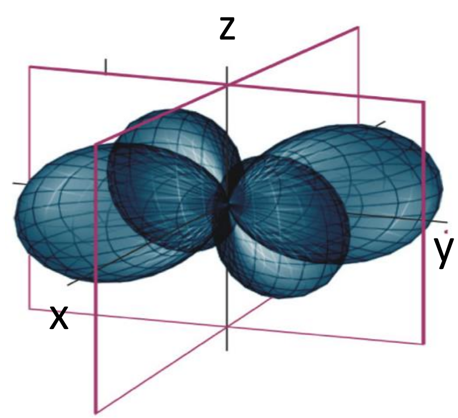
$$\psi_{3d_{x^2-y^2}}=\left\{\frac{1}{81\sqrt{2\pi}}\right\}\left(\frac{z}{a_0}\right)^{\frac{7}{2}}e^{\frac{-zr}{3a_0}}\left((r\sin\theta\cos\phi)^2-(r\sin\theta\sin\phi)^2\right)$$



$$\frac{\psi_{3d_2}-\psi_{3d_{-2}}}{i}=\left\{\frac{1}{81\sqrt{2\pi}}\right\}\left(\frac{z}{a_0}\right)^{\frac{7}{2}}e^{\frac{-zr}{3a_0}}r^2\sin^2\theta\sin 2\phi$$

$$=\left\{\frac{1}{81\sqrt{2\pi}}\right\}\left(\frac{z}{a_0}\right)^{\frac{7}{2}}e^{\frac{-zr}{3a_0}}r^2\sin^2\theta\, 2\sin\phi\cos\phi$$

$$\psi_{3d_{xy}}=\left\{\frac{1}{81\sqrt{2\pi}}\right\}\left(\frac{z}{a_0}\right)^{\frac{7}{2}}e^{\frac{-zr}{3a_0}}(r\sin\theta\cos\phi)(r\sin\theta\sin\phi)$$



Electron spin

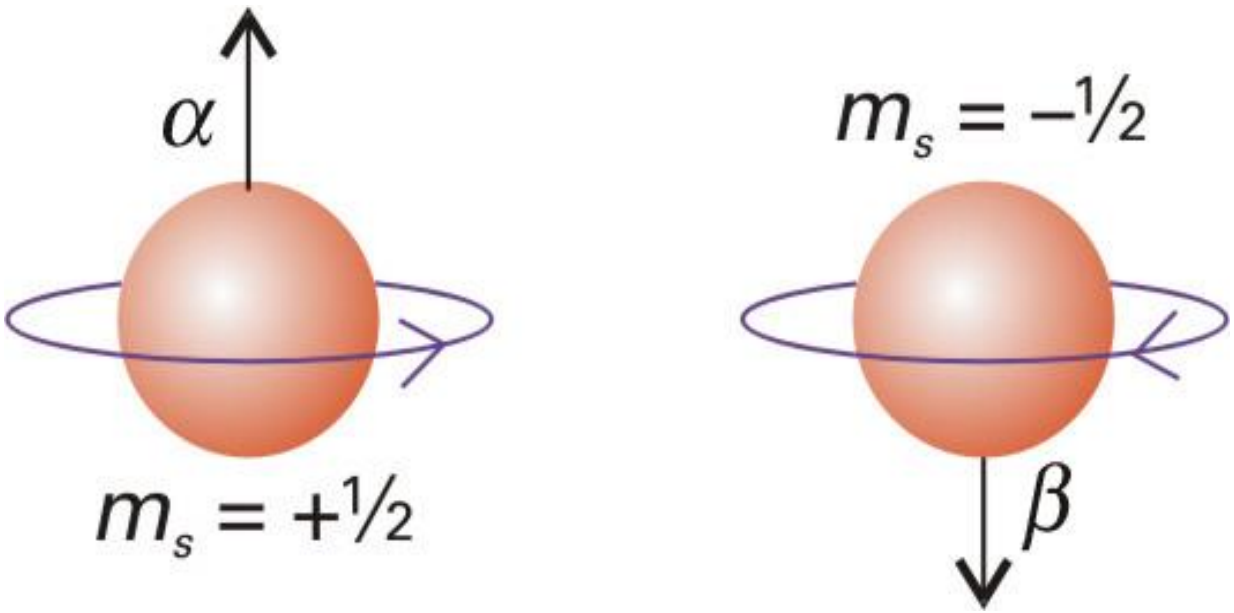
- Spin of an electron is an intrinsic angular momentum, which cannot be changed or eliminated.

- Electron spin is defined by a spin quantum number, *s* with a fixed value of ½ for all electrons.

- The spin can be "clockwise" or "anticlockwise", these states are distinguished by spin magnetic quantum number, m_s, where m_s can take the values = ½ or -½ .

- Experimentally existence of electron spin was first demonstrated by Stern and Gerlach (<u>Self-study</u>)

Spin states of an electron



The state of the electron in the hydrogenic atom is therefore fully specified by the four quantum numbers *n*, *l*, m_l, and m_s.

Spin

- **All fundamental particles have characteristic spin values.**
- **Protons and Neutrons are also spin ½ particles like electrons.**
- **Photons are spin 1 particles.**

All particles may be classified as

- **BOSONS (spin quantum number is zero or a positive integer, ie., s = 0,1,2,…)**

OR

- **FERMIONS (spin quantum number is a positive half-integer, ie., s = 1/2, 3/2,..)**

Spectral transitions

In the one electron atom, spectral line occurs when electron makes a transition between shells with principal quantum numbers n_i and n_f. However, not all transitions between all available orbitals are possible.

For example,

2p to 1s, and 3p to 1s are *allowed* transitions,

2s to 1s or 3d to 1s are *forbidden* transitions.

A statement which specifies the allowed spectroscopic transitions is called a *selection rule*

Selection rule – one electron atom

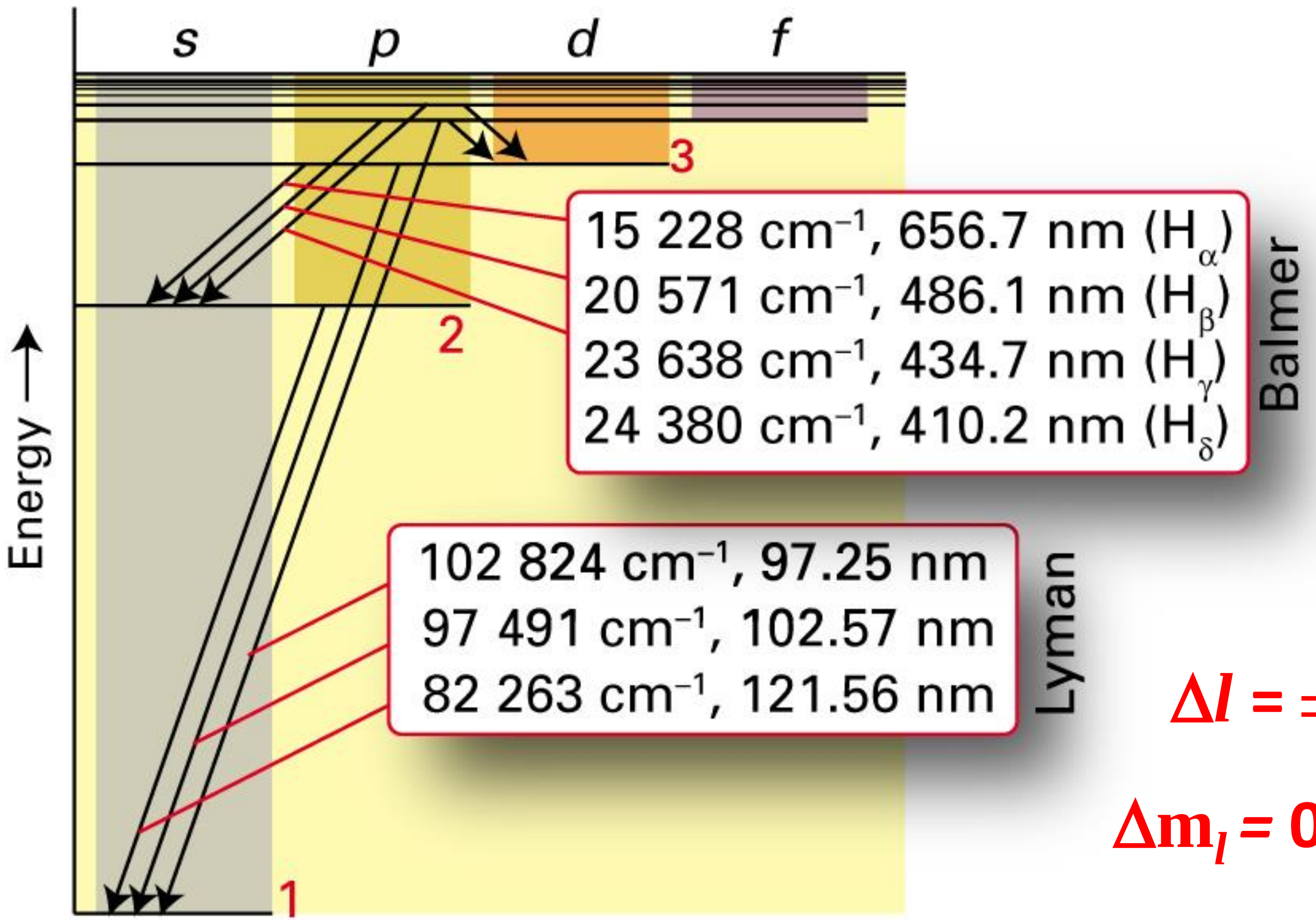
- **For the one electron atom, the selection rule is just the statement of the <u>conservation of angular momentum</u>.**

- **During a transition, angular momentum must be conserved.**
- **When a photon is released (or absorbed) in a transition, the angular momentum of the electron must change in order to compensate for the angular momentum carried away (or lost) by photon.**
- **Selection rule:** $\Delta l = \pm 1$

$$\Delta m_l = 0, \pm 1$$

The principal quantum number can change by any value consistent with the Δl for the transition because n is not directly related to angular momentum.

Grotrian diagram



$\Delta l = \pm 1$

$\Delta m_l = 0, \pm 1$